Moss Pattern Formation on Crushed Basalt in Biosphere 2's LEO Biome

Freddy Aguilar, Sophia Bui, Ava Mohr
The University of Arizona, Department of Mathematics

COLLEGE OF SCIENCE Mathematics

Biosphere 2

Introduction

- Landscape Evolutionary Observatory (LEO) at Biosphere 2 is a long term, environmentally controlled laboratory experiment
  - 30-m x 30-m, 10 slope
- There is an array of more than 1.800 sensors and sampling devices monitoring water, carbon, and energy cycling processes
- Moss is growing in non-uniform patterns on the 1-meter crushed basalt rock surface layer
- Determining ecohydrological and biogeochemical relationship between moss pattern formation and environmental drivers can help facilitate understanding of impact of climate change on water resources and ecosystems in arid environments

Research Question

What are the factors that govern the moss patch formation and spatial patterning on the surface?

Previous Work

Rietkerk model (2002) : three variables (surface water, soil water, plant density), differential flow model that investigates the conditions for formation of spatially nonuniform structures

Klausmeier model (1999): two variables (surface water, plant biomass), models regular patterns on hillsides

Data Available

5 cameras per slope with images taken every hour
- RGB and RGB plus near-infrared images
- Images filtered by quality

Time frame: 2022-10-08 to 2024-01-01
Sensors: carbon dioxide, water content, luminosity, temperature

Methodology

Image Pipeline

Image calibration	Image stitching	Moss pixel calculation

$$\frac{\partial u}{\partial t} = D_u \nabla_u^2 + (r(T,\theta) - \mu(\theta))u(1 - \frac{u}{k})$$

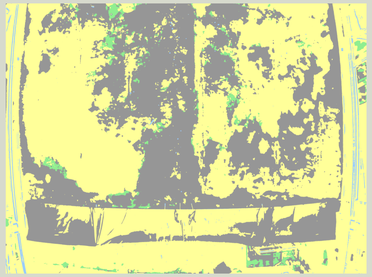
$$r(T,\theta) = r_{max} \exp(-\frac{(T - T_{opt})^2}{2\sigma_T^2}) \frac{\theta}{K_\theta + \theta}$$

$$\mu(\theta) = \mu_{min} + (\mu_{max} - \mu_{min}) \frac{K_\theta}{K_\theta + \theta}$$

$$\frac{\partial \theta}{\partial t} = I(x,y,t) - \beta\mu\theta + D_\theta \nabla^2 \theta + \nabla \cdot (\theta \nabla Z)$$

Text for methodology

Results



Discussion

- Research focused on Long-term environmental research to determine Mars soil growth
- Using the differential equation to determine the influence of moss growth in relation to water content, luminosity, etc.
- Moss area time series indicates correlation with all variables leading to the possibility of promising results on Mars.
- The parameter with the strongest influence:

Conclusion

References

Background

Introduction

- Landscape Evolutionary Observatory (LEO) at Biosphere 2 is a long term, environmentally controlled laboratory experiment
  - 30-m x 30-m, 10 slope
- There is an array of more than 1.800 sensors and sampling devices monitoring water, carbon, and energy cycling processes
- Moss is growing in non-uniform patterns on the 1-meter crushed basalt rock surface layer
- Determining ecohydrological and biogeochemical relationship between moss pattern formation and environmental drivers can help facilitate understanding of impact of climate change on water resources and ecosystems in arid environments

Research Question

What are the factors that govern the moss patch formation and spatial patterning on the surface?

Previous Work

Rietkerk model (2002) : three variables (surface water, soil water, plant density), differential flow model that investigates the conditions for formation of spatially nonuniform structures

Klausmeier model (1999): two variables (surface water, plant biomass), models regular patterns on hillsides

Data Available

5 cameras per slope with images taken every hour
• RGB and RGB plus near-infrared images
• Images filtered by quality
Time frame: 2022-10-08 to 2024-01-01
Sensors: carbon dioxide, water content, luminosity, temperature

Methodology

Results

Discussion

- Research focused on Long-term environmental research to determine Mars soil growth
- Using the differential equation to determine the influence of moss growth in relation to water content, luminosity, etc.
- Moss area time series indicates correlation with all variables leading to the possibility of promising results on Mars.
- The parameter with the strongest influence:

Conclusion

References